

WTW 158 Calculus

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

ASSIGNMENT: SEMESTER TEST 1

April 2021

TIME: 90 minutes to do the assignment. Extra time will be given for downloading and submission.

MARKS: 24

INSTRUCTIONS

1. The assignment consists of 2 pages and 7 questions.
2. Draw a line with a ruler at the end of each question.
3. Write your surname, initials and student number clearly in the beginning of your assignment.
4. **Clearly show all your steps and calculations.**
5. You have 90 minutes to do the assignment.
6. You have 30 minutes to prepare assignment for submittance and to submit.
6. Only handwritten assignments submitted in one pdf file will be accepted.
7. Late assignments will not be accepted.
8. Assignments sent via e-mail will not be accepted.
9. Calculators **may not** be used.

Question 1

Find the equation of the function $f(x) = Ca^x$ that passes through $(1, \frac{3}{2})$ and $(3, \frac{3}{8})$. (2)

Question 2

Sketch the graph of $y = -|\ln(x - 3)|$. Clearly show how you transform the graph of $y = \ln(x)$ to obtain the required graph and indicate all the relevant information on your graph. (2)

Question 3 [10 marks]

Solve for x . Write your answer in interval notation.

3.1 $|\cos x| > \frac{1}{2}, x \in [0, 2\pi]$. (2)

3.2 $\log_2 x + \log_2(x - 1) > 1$. (3)

3.3 $Ce^{bx} - e^{ax} > 0$, where $a > b$ and $C > 0$. (2)

3.4 $\frac{|x-2|-1}{x^2} > 0$ (3)

Question 4

Use the definition of the absolute value function to sketch the graph of

$$y = \sin |x|, x \in [-2\pi, 2\pi].$$

Clearly indicate all the relevant information on the sketch. (2)

Question 5

If $f(x) = x + 4$ and $h(x) = 4x - 1$, find a function g such that $h = g \circ f$. (2)

Question 6

If a bacteria population starts with 100 bacteria and doubles every three hours, then the number of bacteria after t hours is

$$y = f(t) = 100 (2^{\frac{t}{3}}).$$

Find the inverse of this function and explain its meaning. Specify the domain and range of both f and its inverse. (4)

Question 7

Assume that f and g are functions with the same domains. Furthermore, $f \neq 0$ is an odd function and the function

$$y = (fg)(x)$$

is even. Determine if g is even or odd. Explain your answer. (2)